

Generative AI Internship – Task 1

PROJECT: Prompt Engineering Showcase

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Introduction

Prompt Engineering is a key skill in Generative AI that focuses on designing structured instructions to guide AI models toward generating accurate, relevant, and high-quality outputs.

This showcase demonstrates different prompt engineering techniques across multiple domains by comparing basic prompts with optimized prompts. Each optimized prompt includes improved context, role definition, constraints, expected output format, and specific requirements to achieve better AI responses.

The project highlights practical applications of prompt engineering in software development, content creation, data analysis, learning, and creative workflows.

It also demonstrates how modifying prompts with proper context, instructions, and constraints can improve the quality of AI-generated responses.

1. Coding Domain

Prompt 1

Basic Prompt:

Create a website.

Optimized Prompt:

Act as a frontend developer. Create a responsive personal portfolio website using HTML, CSS, and JavaScript. Include sections for About Me, Skills, Projects, and Contact. Use a clean modern design and explain the structure of the code.

Improvement:

Added expert role, technologies, design requirements, and explanation format.

Prompt 2

Basic Prompt:

Fix my code.

Optimized Prompt:

Analyze the following Python code, identify errors, explain the cause of each issue, and provide the corrected version with suggestions for improving code quality and efficiency.

Improvement:

Specified programming language, debugging process, and expected improvements.

Prompt 3

Basic Prompt:

Build an application.

Optimized Prompt:

Act as a software engineer. Develop a task management application using Python. Include features such as adding tasks, deleting tasks, updating status, and saving user data. Explain the implementation step-by-step.

Improvement:

Defined application purpose, features, and development approach.

Prompt 4

Basic Prompt:

Explain Java.

Optimized Prompt:

Explain Java programming concepts for a beginner. Cover variables, data types, loops, classes, inheritance, and object-oriented programming with simple examples.

Improvement:

Defined learning level, topics, and example requirements.

Prompt 5

Basic Prompt:

Create a chatbot.

Optimized Prompt:

Act as an AI developer. Design a simple chatbot solution using Generative AI. Explain the workflow, required technologies, possible use cases, and implementation approach.

Improvement:

Added professional role, technical context, and structured output.

2. Writing Domain

Prompt 6

Basic Prompt:

Write about AI.

Optimized Prompt:

Write a beginner-friendly article explaining Artificial Intelligence, including its applications, benefits, limitations, and real-world examples. Maintain a clear and informative tone.

Improvement:

Added audience, structure, and writing style.

Prompt 7

Basic Prompt:

Write an email.

Optimized Prompt:

Write a professional email requesting internship opportunities from a company. Keep the tone polite, concise, and suitable for a student.

Improvement:

Specified purpose, audience, and communication style.

Prompt 8

Basic Prompt:

Create a resume.

Optimized Prompt:

Create an ATS-friendly resume format for a Computer Science student applying for internships. Include education, technical skills, projects, certifications, and achievements.

Improvement:

Added target audience and required sections.

Prompt 9

Basic Prompt:

Write a LinkedIn post.

Optimized Prompt:

Create a professional LinkedIn post announcing my Generative AI internship experience. Include achievements, learning outcomes, and relevant hashtags.

Improvement:

Defined platform, objective, and professional tone.

Prompt 10

Basic Prompt:

Write a report.

Optimized Prompt:

Create a structured technical report on Generative AI. Include introduction, key concepts, applications, challenges, and conclusion.

Improvement:

Added report format and content expectations.

3. Summarization Domain

Prompt 11

Basic Prompt:

Summarize this.

Optimized Prompt:

Summarize this article into five important bullet points highlighting the main ideas, key findings, and conclusion.

Improvement:

Specified output format and important information.

Prompt 12

Basic Prompt:

Make notes.

Optimized Prompt:

Convert this chapter into simple study notes with headings, key concepts, definitions, and examples for better understanding.

Improvement:

Defined learning format and organization.

Prompt 13

Basic Prompt:

Explain this document.

Optimized Prompt:

Analyze this document and explain the main concepts in simple language suitable for a beginner. Highlight important points and examples.

Improvement:

Added audience level and explanation requirements.

Prompt 14

Basic Prompt:

Summarize a video.

Optimized Prompt:

Summarize this video transcript into key points, important examples, major discussions, and final conclusions.

Improvement:

Specified summary structure and focus areas.

Prompt 15

Basic Prompt:

Shorten this text.

Optimized Prompt:

Rewrite this content into a concise version while maintaining the original meaning, important details, and professional tone.

Improvement:

Added preservation requirements and writing style.

4. Data Analysis Domain

Prompt 16

Basic Prompt:

Analyze data.

Optimized Prompt:

Analyze this dataset and identify important trends, patterns, relationships, and insights. Provide conclusions and recommendations based on the findings.

Improvement:

Defined analysis objectives and expected output.

Prompt 17

Basic Prompt:

Find patterns.

Optimized Prompt:

Analyze this dataset to identify hidden patterns, unusual values, correlations, and possible factors affecting the results.

Improvement:

Specified pattern types and analysis focus.

Prompt 18

Basic Prompt:

Create a report.

Optimized Prompt:

Generate a professional data analysis report including observations, insights, visual representations, and recommendations.

Improvement:

Added report structure and deliverables.

Prompt 19

Basic Prompt:

Explain data.

Optimized Prompt:

Explain this dataset in simple terms and describe what each column represents, its purpose, and how it contributes to analysis.

Improvement:

Defined explanation depth and objective.

Prompt 20

Basic Prompt:

Visualize data.

Optimized Prompt:

Create suitable data visualizations for this dataset and explain which charts are most effective for understanding trends and comparisons.

Improvement:

Added visualization purpose and reasoning.

5. Creative Domain

Prompt 21

Basic Prompt:

Write a story.

Optimized Prompt:

Write a mystery story featuring a young female detective as the main character. Create a suspenseful atmosphere with an unexpected ending.

Improvement:

Added genre, character details, and creative direction.

Prompt 22

Basic Prompt:

Give project ideas.

Optimized Prompt:

Generate 10 innovative project ideas for Computer Science students interested in Artificial Intelligence. Include problem statements and possible solutions.

Improvement:

Defined target audience and output requirements.

Prompt 23

Basic Prompt:

Create a caption.

Optimized Prompt:

Generate creative Instagram captions for a cozy reading and lifestyle account. Keep them aesthetic, engaging, and suitable for a young audience.

Improvement:

Added context, platform, and style.

Prompt 24

Basic Prompt:

Suggest names.

Optimized Prompt:

Suggest unique and memorable names for an AI-based application. Names should be short, professional, creative, and brand-friendly.

Improvement:

Added naming criteria and purpose.

Prompt 25

Basic Prompt:

Create a design idea.

Optimized Prompt:

Generate creative UI/UX design ideas for a modern AI application. Include layout suggestions, user experience improvements, and important features.

Improvement:

Defined design goal, audience, and expected output.

Reusable Prompt Template Library

Coding Template

Act as a software developer. Create [project name] using [technology]. Include [requirements]. Explain the solution step-by-step.

Learning Template

Teach me [topic] as a beginner. Explain concepts clearly with examples and practical applications.

Writing Template

Write a [type of content] about [topic]. Use a [tone] style and target [audience].

Analysis Template

Analyze this [data/document]. Identify key insights, patterns, and provide recommendations.

Conclusion

This task helped me understand the importance of prompt engineering in Generative AI. By creating structured prompts with clear instructions, context, and expected outputs, AI models can generate more accurate, relevant, and useful responses. This project improved my understanding of designing effective prompts for different real-world applications.

Tools Used

- ChatGPT
- Claude
- Microsoft Word